

SMART MULTI-SPECIALITY HOSPITAL NETWORK

R0 (ISP Router) & ISP-Side Switch — VLAN Configuration

VLAN	Switch Port	Connects To	R0 Sub-Interface	R0 IP	Subnet
101	Fa0/2	R1 (HQ Edge)	Gi0/0.101	200.1.1.2	/30
104	Fa0/3	R4 (Diagnostic Edge)	Gi0/0.104	200.1.1.6	/30
106	Fa0/4	R6 (Emergency Edge)	Gi0/0.106	200.1.1.10	/30
109	Fa0/5	R9 (Telemedicine Edge)	Gi0/0.109	200.1.1.14	/30
111	Fa0/6	R11 (Maternity Edge)	Gi0/0.111	200.1.1.18	/30
113	Fa0/7	R13 (Children Care Edge)	Gi0/0.113	200.1.1.22	/30

The trunk port on the ISP switch (Fa0/1, facing R0) carries all six VLANs. Each VLAN maps one-to-one with a dedicated /30 subnet, allowing R0 to reach every Edge router over a single physical link using sub-interfaces (router-on-a-stick).

ISP-Side Switch Configuration

```
en
conf t

vlan 101
vlan 104
vlan 106
vlan 109
vlan 111
vlan 113

int fa0/2
switchport mode access
switchport access vlan 101

int fa0/3
switchport mode access
switchport access vlan 104
int fa0/4
```

```
switchport mode access
switchport access vlan 106
```

```
int fa0/5
switchport mode access
switchport access vlan 109
```

```
int fa0/6
switchport mode access
switchport access vlan 111
```

```
int fa0/7
switchport mode access
switchport access vlan 113
```

```
int fa0/1
switchport mode trunk
switchport trunk allowed vlan 101,104,106,109,111,113
```

R0 (ISP Router) Configuration

```
en
conf t
```

```
int gi0/0
no ip address
no shut
```

```
int gi0/0.101
encapsulation dot1Q 101
ip address 200.1.1.2 255.255.255.252
```

```
int gi0/0.104
encapsulation dot1Q 104
ip address 200.1.1.6 255.255.255.252
```

```
int gi0/0.106
encapsulation dot1Q 106
ip address 200.1.1.10 255.255.255.252
```

```
int gi0/0.109
encapsulation dot1Q 109
ip address 200.1.1.14 255.255.255.252
```

```
int gi0/0.111
encapsulation dot1Q 111
ip address 200.1.1.18 255.255.255.252
```

```
int gi0/0.113
encapsulation dot1Q 113
ip address 200.1.1.22 255.255.255.252
```